TK

Requirements Specification Package

Document	SiamMart Returns — Requirements Package (Output 2)
Customer	SiamMart Co., Ltd. (retail chain, ~120 branches + online)
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Document Control

Version history

Version	Date	Author	Change summary
v0.1	23 Oct 2026	TK BA	Initial statement log and analysis from kickoff notes.
v1.0	29 Oct 2026	TK BA	Complete requirements package for review: scope, TR/NFR/INT, functional spec, user stories, commercial notes.

Author, reviewers & approvers

Role	Name	Responsibility	Signature / date
Author	TK Company — Business Analyst	Authors the requirements package	
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Approver (customer)	K.Somchai — Head of Customer Service	Business sign-off	
Approver (customer)	K.Pim — Finance	Refund / audit sign-off	
Approver (customer IT)	K.Arthit — IT	Integration & security sign-off	

Status of this document

This is a **v1.0 draft for customer review**. It records what the customer said in the kickoff, in a precise, testable form. Where customer statements conflict or are unverified, this document does **not** decide — it raises an `OI-` (Open Issue) or records an `AS-` (Assumption) with an owner. Items marked **OPEN** or **CONFLICT** require a customer decision before design freeze. Section 14 (Commercial Notes) is **internal to TK** and is removed from any customer-facing copy.

> **How to read the IDs.** `BR-` business objective · `BRL-` business rule · `TR-<MOD>-` technical requirement · `NFR-<CAT>-` non-functional requirement · `INT-<SYS>-` integration requirement · `FS-<MOD>-` functional spec · `EP-/US-` epic / user story · `AS-` assumption · `OI-` open issue. All items carry a source reference (S-xx) to the statement log in §2.3.

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1 Introduction

SiamMart is a Thai retail chain of roughly **120 branches** with an online shop (own website plus Shopee and Lazada). Returns and refunds today run on paper, ad-hoc Excel files and a LINE group, and the average refund takes about **14 days** (range 10–21) against a competitor benchmark of **3 days**. Slow refunds were the **#1 complaint** on social media last quarter.

SiamMart wants a single, unified returns and refund platform covering customer self-service, in-store tablet processing, approval workflow, refund posting to SAP, returned-goods disposition, customer notifications, executive dashboards and fraud flagging. This document captures those requirements in a precise, testable, commercially-aware package so developers, testers and SiamMart can agree scope and sign off.

It is derived **only** from the kickoff meeting notes dated 22 Oct 2026 (kept as `input.md`). Every number, rule and integration traces to a source statement; anything unverified is raised as an assumption or an open issue rather than invented.

2 Business Context & Objectives

2.1 The real problem

The pain is not "we have no software" — it is that a return is **an untracked, multi-handoff manual process**: a customer return is written on paper or Excel, photographed into a LINE group, approved in a chat, and refunded manually in SAP once a week. The consequences are measurable:

- **Slow, opaque refunds** — ~14 days, ~4.5× the 3-day competitor benchmark (S-05).
- **No single source of truth** — parallel paper, Excel ("RETURN_MASTER_v7_final_FINAL.xlsx") and LINE trails; no reliable count, reason or store reporting (S-03, S-04).
- **Refund complaints are the top social complaint** — ~2,300 per quarter (unconfirmed, S-06; baseline AS-07).
- **Returned goods pile up behind stores** with no disposition decision — value is lost and stock is mismanaged (S-07).
- **Fraud is invisible** — no way to spot serial returners (S-20).

2.2 SMART business objectives

ID	Objective (specific, measurable, testable)	Baseline → Target	Source
BR-01	Reduce end-to-end refund lead time for approved returns.	~14 days → **≤ 3 business days (p95)**	S-05
BR-02	Reduce refund-related customer complaints captured per quarter.	~2,300 → **–50% within 2 quarters** (baseline AS-07; target to confirm AS-17)	S-06
BR-03	Consolidate every return into one system of record; eliminate paper/Excel/LINE handoffs.	**100% of returns** captured in platform	S-02,S-03,S-04,S-08
BR-04	Track and dispose of every returned item to a final disposition.	**100% of items** with disposition decision	S-07,S-19
BR-05	Surface fraud candidates (serial returners) for CS review.	**Flag rule configurable**; 0 unflagged known cases	S-20
BR-06	Absorb post-11.11 / 12.12 return peaks without degradation.	avg 25k/mo → **4–5× peak (~4,200/day)**	S-22,S-35
BR-07	Give executives visibility of return volume, reasons, store hotspots and fraud.	**Daily-refresh dashboard**	S-17
BR-08	Administer master data, users, roles and configurable business-rule thresholds in a governed, audited way.	All master data & thresholds configurable; **100% changes audit-logged**	S-11,S-12,S-20,S-24

2.3 Statement Log — every customer statement classified with source

Tags: **Goal** · **Pain** · **Req**(requirement) · **Cons**(constraint) · **AS**(assumption) · **Conf**(conflict) · **OQ**(open question) · **Noise**. Source references are line numbers in the kickoff notes.

S	Customer statement (verbatim / paraphrased)	Tag	Traces to
S-01	SiamMart = retail chain ~120 branches + online shop (website + Shopee/Lazada)	Context	BR-01, §3
S-02	"ตอนนี้ลูกค้าคืนของ / ขอ refund วุ่นวายมาก" — returns/refunds are chaotic today	Pain	BR-03
S-03	In-store return: staff write on paper / some branches Excel → photo to LINE group "Return-HQ" → CS at HQ approves in LINE → Finance manual refund in SAP every Friday	Pain	BR-03, BRL-003
S-04	Online return: customer emails returns@siammart.co.th or chats → CS copies into central Excel "RETURN_MASTER_v7_final_FINAL.xlsx" (shared drive)	Pain	BR-03
S-05	Refund avg 14 days (K.Pim: 10–21 days depends); competitor 3 days	Pain/Metric	BR-01
S-06	Slow refund = #1 complaint on social last quarter (~2,300 complaints, number needs confirm)	Pain/AS	BR-02, AS-07
S-07	Returned goods pile behind store; unclear send to DC / discard / markdown	Pain/Req	BR-04, BRL-009
S-08	"อยากได้ระบบ return ที่ทุกอย่างอยู่ที่เดียว" — want one return system, everything in one place	Goal	BR-03
S-09	Customer can request return via web/app (no app? K.Somchai: "LINE OA ก็ได้")	Req	TR-RET-01, AS-05
S-10	Store staff use tablet to scan receipt barcode → system pulls order	Req	TR-STORE-01, INT-POS-01
S-11	If price < 500 THB auto-approve (K.Somchai) / K.Pim: "must pass finance every item per audit"	**Conf**	BRL-002, OI-01
S-12	If refund > 5,000 THB, manager must approve	Req	BRL-003
S-13	Refund to credit card / PromptPay / store credit (store credit +5% bonus? marketing idea, not confirmed)	Req/AS	BRL-005, AS-04
S-14	Must integrate SAP (finance) — SAP ECC 6.0, has RFC but don't want anyone touching directly	Req/Cons	INT-SAP-01
S-15	Must integrate POS (vendor "RetailPro"? or in-house? K.Arthit not sure)	Req/OQ	INT-POS-01, OI-08
S-16	Must integrate Shopee/Lazada returns too? → phase 2 maybe	OQ/Scope	§3 OOS-1, OI-09
S-17	Dashboard for executives: # returns, reasons, which branch returns most, fraud?	Req	TR-DASH-01, BR-07
S-18	Notify customer every step (SMS? LINE? email?) "like Kerry tracking"	Req	TR-NOTIF-01
S-19	Returned goods need a flow: restock / send back to supplier / write-off	Req	TR-INV-01, BRL-009
S-20	Fraud: some customers return every week — system should flag	Req	TR-FRAUD-01, BRL-007

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2.3 Statement Log (continued)

S	Customer statement (verbatim / paraphrased)	Tag	Traces to
S-21	"ต้องใช้ง่ายมาก พนักงานบางคนอายุเยอะ" — must be very easy, some staff are older	Cons/NFR	NFR-USA-01
S-22	Must handle peak after 11.11 / 12.12 — returns spike 4–5×	Cons/NFR	NFR-SCAL-01, BR-06
S-23	Product photos must be kept (how many years? audit says 7 years? not sure)	Req/OQ	BRL-008, AS-03, OI-06
S-24	Customer data → PDPA!! (K.Arthit concern)	Cons	NFR-PRIV-01
S-25	Language TH/EN	Req	NFR-LOC-01
S-26	Budget: "ไม่เกิน 8 ล้าน phase 1" (K.Somchai) — Finance not fully agreeing	Cons/Conf	§12, OI-11, §14
S-27	Go-live before 11.11 next year (Nov 2027) "ต้องทัน"	Cons	§12, §14
S-28	Want to pilot 10 Bangkok branches first	Cons/Scope	§3 Phasing
S-29	IT SiamMart only 4 people → want TK to host + support (MA?)	Cons/Req	NFR-SUPP-01
S-30	Don't want to buy expensive licences	Cons	§14 (commercial)
S-31	Return policy: 7 days? 14 days? different by category (electronics 7, other 30?) → ask K.Nok for policy doc	OQ	BRL-001, OI-02
S-32	Who owns this process? CS or Store Ops?	OQ	OI-03
S-33	Partial return (return some items in an order) allowed? → probably yes	OQ/Req	BRL-006, OI-10
S-34	Exchange (change item) in scope?	OQ	BRL-010, OI-05
S-35	Return volume / month: K.Nok ~18k, K.Somchai ~25k ???	**Conf(fig.)**	AS-01, OI-04
S-36	Old Excel system — migrate? "ย้อนหลัง 1 ปีพอ" (last 1 year is enough)	AS	AS-02, OI-12
S-37	SSO with SiamMart AD?	OQ	INT-AD-01, OI-07
S-38	K.Somchai liked competitor's "return QR code" (customer gets QR to present in store)	Req(nice)	TR-RET-05
S-39	K.Pim wants daily refund report export to Excel	Req	TR-DASH-04
S-40	Someone mentioned "AI checking item condition photos" — nice to have, ignore for now?	Noise/OOS	§3 OOS-4
S-41	Next meeting: walk-through store process at Bangna branch, date TBC	Noise/Next	§12 dependency

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3 Scope

3.1 In scope — Phase 1 (pilot 10 Bangkok branches → national rollout)

#	In-scope capability	Module	MoSCoW	Source
IS-1	Customer self-service return request via web portal and LINE OA	RET	MUST	S-09
IS-2	In-store return processing on tablet with receipt barcode scan	STORE	MUST	S-10
IS-3	Approval workflow (rules engine, manager threshold, finance visibility)	APPR	MUST	S-11,S-12
IS-4	Refund posting to SAP ECC 6.0 via TK-Connect (idempotent, reconciled)	REF	MUST	S-14
IS-5	Returned-goods disposition: restock / return-to-supplier / write-off	INV	MUST	S-19
IS-6	Customer notifications at each status step (SMS/LINE/email)	NOTIF	MUST	S-18
IS-7	Executive dashboard (volume, reason, store, lead time, fraud)	DASH	MUST	S-17
IS-8	Fraud flagging of serial returners (configurable thresholds)	FRAUD	SHOULD	S-20
IS-9	POS order retrieval integration for store scan	POS	MUST	S-10,S-15
IS-10	Administration: users, roles, stores, categories, reason codes, thresholds	ADMIN	MUST	S-21
IS-11	Daily refund report export to Excel for Finance	DASH	SHOULD	S-39
IS-12	TH/EN localization, PDPA-compliant handling of customer data & photos	ALL	MUST	S-24,S-25
IS-13	Return QR code for in-store drop-off (customer presents QR)	RET	COULD	S-38
IS-14	1-year historical data migration from existing Excel master	ADMIN	SHOULD	S-36

3.2 Out of scope — explicitly excluded, with reason & routing

#	Out-of-scope item	Reason	Routed to
OOS-1	Shopee / Lazada marketplace return integration	Customer said "phase 2 maybe"; not Phase-1 priority (S-16). Deferred.	Phase 2 / OI-09
OOS-2	Native iOS/Android mobile app	No app exists; K.Somchai accepted LINE OA as the customer channel (S-09, AS-05).	Future / AS-05
OOS-3	Exchange (change-of-goods) workflow	Scope not confirmed (S-34); kept out of Phase 1 pending OI-05.	OI-05
OOS-4	AI photo-based item-condition detection	Explicitly "nice to have, ignore for now" (S-40).	Future / §14 upsell
OOS-5	Store-credit +5% bonus programme	Marketing idea, not confirmed (S-13, AS-04). Store credit supported without the bonus.	Marketing / AS-04
OOS-6	Full AP/GL accounting, procurement, warehouse management	Belongs to SAP / ERP; platform posts refunds to SAP only.	SAP / tk-solution-architecture
OOS-7	Hardware procurement for 120 stores (tablets) if not already owned	Device availability unconfirmed; commercial item (§14).	§14 / customer IT

3.3 Phasing

Phase	Content	Exit criterion
Phase 1 (pilot)	MUST + SHOULD scope, 10 Bangkok branches, core return→refund→disposition→notify→dashboard.	Pilot stores live; refund p95 ≤ 3 days
Phase 1 rollout	Remaining ~110 branches; scaled capacity for 11.11/12.12 peak.	National go-live before 11 Nov 2027
Phase 2	Marketplace (Shopee/Lazada) returns, exchange flow, AI condition check, mobile app, store-credit programme.	Separate CR / budget

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4 Stakeholders & RACI

4.1 Stakeholders

Name	Role	Interest / concern in this project
K.Somchai	Head of Customer Service (SiamMart)	Business sponsor; wants fast, unified returns; proposes <500 THB auto-approval; budget owner (≤8M).
K.Nok	Store Operations (SiamMart)	Store process, return policy doc, store-side usability, disposition.
K.Pim	Finance (SiamMart)	Refund control & audit; wants every refund through finance; daily Excel report.
K.Arthit	IT (SiamMart)	SAP/POS/AD integration & security; PDPA; small team (4) needing TK hosting/MA.
TK BA / SA	TK Company delivery	Author requirements & architecture; ensure testable, compliant, on-budget.
Customers	SiamMart shoppers (in-store + online)	Fast, transparent refunds; easy self-service; QR drop-off.

4.2 RACI (key activities)

R = Responsible · A = Accountable · C = Consulted · I = Informed

Activity	K.Somchai	K.Nok	K.Pim	K.Arthit	TK BA	TK SA
Requirement sign-off	A	C	C	C	R	I
Return policy definition	C	A/R	I	I	C	I
Refund/approval rule decision (OI-01)	A	I	A	I	C	I
SAP / POS integration	I	C	C	A	C	R
PDPA / security	I	I	C	A	C	R
UAT & acceptance	A	R	C	C	C	I
Budget approval	A	I	C	I	I	I
Hosting / MA decision	C	I	I	A	I	R
Go-live readiness	A	R	C	R	I	R

> Process ownership (CS vs Store Ops) is **not yet decided** — see `OI-03`. The RACI above assumes joint accountability until that is resolved; it will be re-baselined once the owner is named.

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5 Business Rules (BRL)

Each rule = condition → action, with priority, status and source. Rules marked **CONFLICT** or **OPEN** must not be implemented until the referenced OI is resolved.

ID	Rule (condition → action)	Priority	Status	Source
BRL-001	A return is eligible only if the request is within the category return window (electronics 7 days; other categories 30 days — policy TBC) from delivery/purchase date.	MUST	**OPEN** (OI-02)	S-31
BRL-002	If a refund line value < 500 THB → auto-approve without manual CS review (K.Somchai).	MUST	**CONFLICT** (OI-01)	S-11
BRL-003	If total refund > 5,000 THB → require store/branch manager approval before refund.	MUST	**Confirmed**	S-12
BRL-004	Every refund must pass through Finance approval/visibility for audit (K.Pim).	MUST	**CONFLICT** (OI-01)	S-11
BRL-005	Refund is returned to the original payment method (card / PromptPay); store credit is offered as an alternative; cash refunds via PromptPay.	MUST	**Confirmed**	S-13
BRL-006	Partial return — return a subset of line items in an order — is allowed; refund is calculated per returned line.	MUST	**OPEN** (OI-10, likely yes)	S-33
BRL-007	A customer exceeding a configurable threshold of returns within a rolling period (e.g. ≥ N returns / 90 days) → flag for CS fraud review.	SHOULD	**OPEN** (threshold TBD)	S-20
BRL-008	At least one photograph of the returned item(s) is mandatory at intake (store or online).	MUST	**Confirmed**	S-23
BRL-009	Each returned item must be routed to exactly one disposition: RESTOCK (sellable) / RETURN_TO_SUPPLIER (defective, in supplier window) / WRITE_OFF (non-sellable).	MUST	**Confirmed**	S-19,S-07
BRL-010	Exchange (change-of-goods) is *not* part of the return refund flow; handled separately if approved.	N/A p1	**OPEN** (OI-05)	S-34
BRL-011	Once approved, the refund must be completed within ≤ 3 business days (SLA).	MUST	**Confirmed (BR-01)**	S-05
BRL-012	The platform must sustain return peaks of 4–5× average volume (post-11.11 / 12.12).	MUST	**Confirmed**	S-22
BRL-013	In-store return requires a receipt/barcode (or order lookup); online return requires the order number/reference.	MUST	**Confirmed**	S-03,S-10
BRL-014	Refund posting to SAP must be idempotent — a refund may not be posted twice for the same return.	MUST	**Confirmed**	S-14

5.1 Rule engine decision summary (approval routing)

refund_total = Σ returned line values IF refund_total > 5,000 THB → APPROVAL = Manager (+ Finance per BRL-004) ELSE IF refund_total < 500 THB → APPROVAL = Auto-approve [PENDING OI-01 — may become Finance] ELSE (500 – 5,000 THB) → APPROVAL = CS review (+ Finance per BRL-004) ALWAYS → Finance visibility & audit trail (BRL-004 / OI-01) Status: the 500-THB branch is BLOCKED until OI-01 is decided.

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6 Technical Requirements (TR)

Each TR is one testable "The system shall …" statement with ID, MoSCoW priority, source and acceptance criterion.

6.1 Module RET — Customer Return Request

ID	Requirement ("The system shall …")	Priority	Source	Acceptance (testable)
TR-RET-01	…let a registered customer submit a return request via web portal and LINE OA, selecting an order and one or more items.	MUST	S-09	Request created & Return ID issued from both channels.
TR-RET-02	…validate order & item eligibility against the return-window rule (BRL-001) and current return status.	MUST	S-31	Out-of-window item is blocked with reason.
TR-RET-03	…capture a reason code and require ≥1 photo per returned item (BRL-008).	MUST	S-23	Submit blocked until photo + reason present.
TR-RET-04	…support partial return of selected line items within one order (BRL-006).	MUST	S-33	Subset selection refunds only returned lines.
TR-RET-05	…generate a unique Return ID and an optional QR code the customer presents in store (IS-13).	COULD	S-38	QR resolves to the correct open return.
TR-RET-06	…present all customer-facing screens in Thai and English (NFR-LOC-01).	MUST	S-25	Language toggle switches all labels.

6.2 Module STORE — In-store Return Processing

ID	Requirement	Priority	Source	Acceptance
TR-STORE-01	…let store staff scan a receipt barcode on a tablet to retrieve the matching order via the POS integration.	MUST	S-10	Scan returns the correct order within 2s (p95).
TR-STORE-02	…create a return case from the scanned order with items pre-filled.	MUST	S-10	Items match order; staff confirms.
TR-STORE-03	…capture a condition grade and photos per item at intake.	MUST	S-23	Item cannot proceed without grade + photo.
TR-STORE-04	…complete order lookup in ≤ 5 taps/steps on a tablet (usability for older staff).	MUST	S-21	Usability test: ≤5 taps, ≤30s task.

6.3 Module APPR — Approval Workflow

ID	Requirement	Priority	Source	Acceptance
TR-APPR-01	…route each refund through a configurable rules engine implementing BRL-002/003/004.	MUST	S-11,S-12	Routing matches rule table; 500-THB branch disabled until OI-01.
TR-APPR-02	…require manager approval for any refund > 5,000 THB (BRL-003).	MUST	S-12	>5,000 cannot refund without manager.
TR-APPR-03	…present an approval queue with SLA timers and priority ordering.	MUST	S-11	Queue shows age & SLA status.
TR-APPR-04	…record approver identity, timestamp, decision and reason for audit (NFR-AUDIT-01).	MUST	S-11,S-24	Every decision has an immutable audit row.

6.4 Module REF — Refund Execution

ID	Requirement	Priority	Source	Acceptance
TR-REF-01	…post approved refunds to SAP ECC 6.0 via the TK-Connect integration.	MUST	S-14	Refund document created in SAP.
TR-REF-02	…record refund channel (card/PromptPay/store credit) and reference number.	MUST	S-13	Channel + reference stored per refund.
TR-REF-03	…enforce idempotent refund posting — reject duplicate posting for a return (BRL-014).	MUST	S-14	Second post for same return is rejected.
TR-REF-04	…reconcile refund status against SAP daily and surface mismatches.	MUST	S-14	Reconciliation report runs daily; mismatches listed.

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6.5 Module INV — Returned Goods Disposition

ID	Requirement	Priority	Source	Acceptance
TR-INV-01	…record exactly one disposition decision per returned item (RESTOCK / RETURN_TO_SUPPLIER / WRITE_OFF) per BRL-009.	MUST	S-19	Item cannot close without a disposition.
TR-INV-02	…generate return-to-DC / return-to-supplier documents from disposition data.	SHOULD	S-19	Document generated with item list.
TR-INV-03	…update item/disposition status as goods move through the disposition flow.	MUST	S-19	Status transitions tracked per item.

6.6 Module NOTIF — Customer Notifications

ID	Requirement	Priority	Source	Acceptance
TR-NOTIF-01	…notify the customer at each status change via their chosen channel(s): SMS / LINE / email (S-18).	MUST	S-18	Each status triggers a notification on chosen channel.
TR-NOTIF-02	…use TH/EN notification templates and record every send for audit.	MUST	S-18,S-25	Template per language; send log row per message.

6.7 Module DASH — Dashboard & Reporting

ID	Requirement	Priority	Source	Acceptance
TR-DASH-01	…show executive KPIs: return count, reason breakdown, store ranking, refund lead time, fraud flags.	MUST	S-17	All listed KPIs render with daily data.
TR-DASH-02	…filter dashboards by date range, store, category and channel.	MUST	S-17	Filters apply correctly to KPIs.
TR-DASH-03	…export dashboard data to Excel/PDF.	SHOULD	S-17	Export matches on-screen figures.
TR-DASH-04	…produce the daily refund report export (Excel) for Finance (K.Pim).	SHOULD	S-39	Daily file generated & downloadable.

6.8 Module FRAUD — Fraud Flagging

ID	Requirement	Priority	Source	Acceptance
TR-FRAUD-01	…flag customers exceeding the configurable return threshold (BRL-007) and surface them to CS.	SHOULD	S-20	Customer over threshold appears in fraud list.
TR-FRAUD-02	…allow admins to configure the flag thresholds.	SHOULD	S-20	Threshold change alters flagging behaviour.

6.9 Module ADMIN — Administration

ID	Requirement	Priority	Source	Acceptance
TR-ADMIN-01	…let admins manage users, roles, stores, categories, reason codes and rule thresholds.	MUST	S-21	Each master-data type is CRUD-able.
TR-ADMIN-02	…enforce role-based access control on every screen and action (NFR-SEC-01).	MUST	S-24	Unauthorised role cannot access restricted action.
TR-ADMIN-03	…log all configuration changes with before/after values (NFR-AUDIT-01).	MUST	S-24	Config changes appear in audit log.

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7 Non-Functional Requirements (NFR)

Full NFR checklist — every row covered or marked N/A with reason. Banned words (fast, user-friendly…) replaced with numbers.

ID	Category	Requirement (measurable)	Source
NFR-PERF-01	Performance	p95 page load ≤ 2s; p95 API response ≤ 500ms; receipt scan lookup ≤ 2s (TR-STORE-01).	S-21
NFR-SCAL-01	Scalability (peak)	Sustain 4–5× baseline. Baseline sized at 25,000 returns/month (~833/day); peak ≈ **4,200 returns/day**; capacity provisioned with headroom to 6× (≈5,000/day) (AS-01).	S-22,S-35
NFR-AVAIL-01	Availability & RTO/RPO	99.5% monthly availability (business hours weighted); RTO ≤ 4h; RPO ≤ 1h.	Derived — business-critical refund processing (BR-01, BR-03)
NFR-SEC-01	Security & access control	RBAC; TLS 1.2+ in transit; AES-256 at rest; SSO via SiamMart AD where available (INT-AD-01 / OI-07); least-privilege roles.	S-24,S-37
NFR-PRIV-01	Privacy / PDPA	PDPA-compliant: capture lawful basis & consent, purpose limitation, PII masking in non-production & in dashboards, retention limits and a right-to-erasure workflow (see §7.2).	S-24
NFR-AUDIT-01	Audit logging	Immutable audit trail of every approval, refund, disposition and config change; retained 7 years; tamper-evident.	S-23,S-24
NFR-USA-01	Usability & accessibility	Core store task ≤ 5 taps / ≤ 30s; training ≤ 30 min; large touch targets (≥44px) for older staff; WCAG 2.1 AA for customer portal.	S-21
NFR-LOC-01	Localisation	Full Thai and English UI, notifications and reports; Thai fonts (Sarabun); Thai date (พ.ศ./ค.ศ.) and THB currency formats.	S-25
NFR-DATA-01	Data retention & archiving	Return/refund records & photos retained **7 years** (AS-03, pending audit confirm OI-06); historical migration covers **last 1 year** of Excel data (AS-02).	S-23,S-36
NFR-SUPP-01	Supportability	TK-hosted & managed (SaaS) with monitoring, alerting and MA support, because SiamMart IT has only 4 staff (S-29); documented runbooks & SLA.	S-29
NFR-DEV-01	Browser/device support	Latest 2 versions of Chrome/Edge/Safari; Android & iPad tablets for store; responsive customer web; LINE OA in-app browser supported.	S-09,S-10

7.2 PDPA data inventory (legal basis · retention · masking)

Data field	Classification	Legal basis	Retention	Masking / handling
Customer name, phone, email	Personal (PII)	Contract / consent	7 yrs	Masked in dashboards; full view only to authorised CS/Finance.
Payment / card last-4, PromptPay ref	Sensitive-financial	Contract / legal obligation	7 yrs	Never store full PAN; tokenise; mask to last 4.
Return reason, order history	Personal	Contract / legitimate interest	7 yrs	Aggregate for analytics; access-controlled.
Item condition photos	Personal (may include people)	Consent	7 yrs (AS-03)	Encrypted at rest; access logged; auto-purge at retention end.
Fraud flags / scores	Derived personal	Legitimate interest	7 yrs	Restricted to CS fraud team; explainable to data subject on request.

> **PDPA controls:** consent capture at first data entry; data-subject access & erasure request workflow; processing records (ROPA); breach-notification runbook; Data Processing Agreement between SiamMart (controller) and TK (processor). Retention periods depend on OI-06 confirmation.

8 Integration Requirements (INT)

Per TK architecture rules, channels never touch a database and SAP/POS/marketplace traffic only flows via **TK-Connect**.

ID	System	Direction	Protocol	Trigger / frequency	Data objects · volume · error handling · owner
INT-SAP-01	SAP ECC 6.0 (Finance)	Outbound (post refund); inbound (status)	RFC via TK-Connect	On approval + daily batch (Fri legacy → continuous)	**Data:** refund document, amount, channel, return ref. **Vol:** ~refunds/day at peak. **Errors:** retry w/ backoff, dead-letter, daily reconciliation (TR-REF-04). **Owner:** K.Arthit. **Note:** no direct RFC from platform — only TK-Connect (S-14).
INT-POS-01	POS (vendor TBD: RetailPro / in-house)	Inbound (order lookup)	API / file via TK-Connect	Real-time on receipt scan	**Data:** order header, lines, prices, tender. **Vol:** per scan. **Errors:** fallback manual order search; timeout 2s. **Owner:** K.Arthit. **Open:** vendor unknown — OI-08.
INT-PAY-01	Payment gateway / bank	Outbound (card/PromptPay refund)	API	Per refund	**Data:** txn ref, amount, destination. **Errors:** retry, reconcile with SAP. **Owner:** K.Pim. (Gateway assumed — AS-16.)
INT-SMS-01	SMS gateway	Outbound	HTTP API	Per status event	**Data:** phone, template, vars. **Errors:** retry ≤3, log failures. **Owner:** TK.
INT-LINE-01	LINE OA / Messaging API	Bi-directional	LINE Messaging API	Per event + intake	**Data:** LINE user id, template, return intake. **Errors:** retry, fallback email. **Owner:** TK.
INT-EMAIL-01	Email (SMTP)	Outbound	SMTP	Per event + receipts	**Data:** email, template. **Errors:** queue+retry. **Owner:** TK.
INT-AD-01	Active Directory (SiamMart)	Inbound (auth)	LDAP / SAML	On login	**Data:** identity, groups → RBAC. **Owner:** K.Arthit. **Open:** AD availability — OI-07.
INT-MKT-01	Shopee / Lazada	—	—	—	**Out of scope Phase 1** (OOS-1 / OI-09). Route to Phase 2.

8.1 Integration error-handling standard

- **Idempotency:** every outbound money message carries a unique idempotency key (return ref + attempt) — SAP refuses duplicates (TR-REF-03).
- **Retry:** exponential backoff up to 5 attempts, then dead-letter queue with alert.
- **Reconciliation:** daily automated reconciliation of platform refunds vs SAP postings; mismatches raised to Finance.
- **Observability:** every integration call logged with correlation id; dashboards for failed messages.

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9 Functional Specification (FS)

One `FS-<MOD>-nn` per module. Each covers purpose, actors, pre/post-conditions, main & alternative flows, BRL refs, a screen/field table, status model, notifications, reports and linked TR/US.

9.0 Return-case status model (shared)

DRAFT → SUBMITTED → IN_REVIEW → (APPROVED | REJECTED | FRAUD_HOLD) APPROVED → REFUND_PENDING → REFUNDED → CLOSED any pre-terminal state → CANCELLED (by customer before approval / staff) Item-level (parallel): PENDING_DISPOSITION → RESTOCKED | RETURNED_SUPPLIER | WRITTEN_OFF Refund posting (parallel): PENDING_SAP → POSTED | POST_FAILED (retry/reconcile)

State	Meaning	Allowed next states
`DRAFT`	Customer started a return in portal/LINE	SUBMITTED, CANCELLED
`SUBMITTED`	Submitted with reason + photos	IN_REVIEW, CANCELLED
`IN_REVIEW`	Awaiting CS / manager / finance decision	APPROVED, REJECTED, FRAUD_HOLD
`FRAUD_HOLD`	Flagged for fraud review (BRL-007)	IN_REVIEW, REJECTED
`APPROVED`	Decision approved; refund queued	REFUND_PENDING
`REJECTED`	Return declined (reason recorded)	CLOSED
`REFUND_PENDING`	Refund sent/awaiting SAP + payment	REFUNDED, POST_FAILED
`REFUNDED`	Money returned via channel	CLOSED
`CLOSED`	Return fully complete (incl. disposition)	—
`CANCELLED`	Withdrawn before approval	—

9.1 FS-RET-01 — Customer Return Request

Field	Type	Mandatory	Validation / source
Order / reference	Lookup	Yes	Must resolve to a valid order (POS/SAP/online).
Items to return	Multi-select	Yes	Subset of order lines (BRL-006).
Reason code	Dropdown	Yes	From admin master list.
Photos	Upload	Yes (≥1/item)	Image types, size limit (BRL-008).
Preferred refund method	Radio	Yes	Card / PromptPay / store credit (BRL-005).
Notification channel	Checkbox	Yes	SMS / LINE / email.

Pre-conditions: Customer is authenticated and has ≥1 eligible order within the return window. **Post-conditions:** A return case exists in SUBMITTED with reason, photos and refund method; confirmation notification sent. **Actors:** Customer. **Main flow:** select order → select items → reason → photos → method → submit → Return ID issued (SUBMITTED). **Alt/exception:** ineligible item blocked; photo missing blocked; unknown order → contact CS. **BRL:** 001,005,006,008. **TR:** RET-01..06. **US:** US-001, US-002.

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9.2 FS-STORE-01 — In-store Return Processing

Field	Type	Mandatory	Validation / source
Receipt barcode	Scan	Yes	Resolves order via INT-POS-01.
Order lines to return	Multi-select	Yes	Pre-filled from order.
Condition grade	Dropdown	Yes	New / Good / Damaged / Unsellable.
Item photos	Camera	Yes	≥1 per item (BRL-008).
Disposition intent	Dropdown	No	Restock / supplier / write-off (BRL-009).

Pre-conditions: Staff is authenticated at a store; a valid receipt/order exists. **Post-conditions:** A return case is created (SUBMITTED/IN_REVIEW) with item condition grades and photos; disposition intent recorded. **Actors:** Store staff. **Main flow:** scan receipt → order pulled → select items → grade + photo → submit case. **Alt/exception:** POS lookup fails → manual order search; unknown receipt → deny with message. **BRL:** 008,009,013. **TR:** STORE-01..04, INT-POS-01. **US:** US-004, US-005.

9.3 FS-APPR-01 — Approval Workflow

Field	Type	Mandatory	Validation / source
Case summary & amount	Read-only	—	From return case.
Decision	Approve / Reject	Yes	Role-gated (CS / Manager / Finance).
Decision reason	Text	Yes (on reject)	Free text + reason code.
Approver identity	System	—	From AD session.

Pre-conditions: A return case is in IN_REVIEW with a computed refund total. **Post-conditions:** Case transitions to APPROVED or REJECTED; an immutable audit row is written; approved cases are queued for refund. **Actors:** CS staff, Manager, Finance. **Main flow:** case enters queue → assigned approver reviews → approve/reject → audit row written → on approval, refund queued. **Alt/exception:** >5,000 requires manager (BRL-003); <500 auto branch blocked pending OI-01; all refunds visible to Finance (BRL-004). **BRL:** 002,003,004,011. **TR:** APPR-01..04. **US:** US-006, US-007, US-008.

9.4 FS-REF-01 — Refund Execution

Field	Type	Mandatory	Validation / source
Approved amount	Read-only	—	From approval.
Refund channel	Read-only	—	From return request / original tender.
Destination reference	Text	Yes	Card txn / PromptPay id / store-credit wallet.
SAP document no.	System	—	Returned by INT-SAP-01.

Pre-conditions: Case is APPROVED with a valid refund destination and amount. **Post-conditions:** Refund posted to SAP (idempotent), channel/reference recorded, case = REFUNDED, customer notified. **Actors:** Finance, system. **Main flow:** approved refund → post to SAP (idempotent) → record channel/ref → REFUNDED → notify customer. **Alt/exception:** SAP error → POST_FAILED → retry/alert; duplicate blocked (BRL-014); daily reconciliation. **BRL:** 005,011,014. **TR:** REF-01..04, INT-SAP-01, INT-PAY-01. **US:** US-009, US-010.

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9.5 FS-INV-01 — Returned Goods Disposition

Field	Type	Mandatory	Validation / source
Disposition decision	Dropdown	Yes	Restock / Return to supplier / Write-off (BRL-009).
Disposition note	Text	No	Reason / reference.
Return-to-DC / supplier doc	Generated	—	From item list (TR-INV-02).

Pre-conditions: Returned item is physically received and linked to a return case. **Post-conditions:** Exactly one disposition recorded per item; return-to-DC/supplier document generated; item status advanced. **Actors:** Store staff, Warehouse/DC. **Main flow:** received item → assign disposition → generate doc → status updated. **Alt/exception:** item cannot close without a disposition. **BRL:** 009. **TR:** INV-01..03. **US:** US-011.

9.6 FS-NOTIF-01 — Notifications

Event	Channel	Recipient	Template / timing
Return submitted	SMS/LINE/email	Customer	"Return received" — immediate
Approved / Rejected	SMS/LINE/email	Customer	Decision + next step — on decision
Refund posted	SMS/LINE/email	Customer	"Refund on its way" + amount — on post
Refund completed	SMS/LINE/email	Customer	"Refund completed" — on settle
Fraud hold	Internal	CS fraud team	Alert — on flag

Pre-conditions: A status-change event occurred and the customer has a notification preference on file. **Post-conditions:** Message delivered on the chosen channel and logged; fallback attempted on failure. **TR:** NOTIF-01..02. **US:** US-012. Templates in TH & EN.

9.7 FS-DASH-01 — Dashboard & Reports

KPI / report	Formula / definition	Refresh	Audience
Return volume	COUNT(returns) by day/week/month	Daily	Exec / CS
Reason breakdown	% returns by reason code	Daily	Exec / Ops
Store ranking	Returns per store, top-N highest	Daily	Ops
Refund lead time	median & p95 (approval → refunded)	Daily	Exec / Finance
Fraud flags	COUNT(flagged customers/returns)	Daily	CS fraud
Daily refund export	Excel of refunds posted today (TR-DASH-04)	Daily	Finance (K.Pim)

Pre-conditions: Return/refund data is available for the selected period and the user has report permission. **Post-conditions:** KPIs and export rendered matching source data. **TR:** DASH-01..04. **US:** US-013, US-010. Data refreshed daily; export matches on-screen.

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9.8 FS-FRAUD-01 — Fraud Flagging

Field	Type	Mandatory	Validation / source
Customer return-frequency history	Read-only (system)	—	Computed from return records over the rolling period (BRL-007).
Fraud threshold (N returns / period)	Config (from ADMIN)	Yes	Managed in FS-ADMIN-01; default TBD (BRL-007).
Flag decision	System	—	If frequency > threshold → set FRAUD_HOLD.
CS review outcome	CS action	Yes	Release (→ IN_REVIEW) or Reject (→ REJECTED); note required.

Pre-conditions: Return/customer history exists and the fraud threshold is configured in ADMIN. **Post-conditions:** An over-threshold customer is placed in FRAUD_HOLD, CS is notified, and the case is blocked from auto-approval/refund until reviewed. **Actors:** CS fraud analyst, system. **Main flow:** evaluate return frequency per customer against threshold (BRL-007) → flag → state = FRAUD_HOLD → notify CS → CS reviews → release (IN_REVIEW) or reject (REJECTED). **Alt/exception:** below threshold → no action; false positive → CS releases to IN_REVIEW. **Uses status model:** enters FRAUD_HOLD; exits to IN_REVIEW/REJECTED (§9.0). **BRL:** 007. **TR:** FRAUD-01..02. **US:** US-014.

9.9 FS-ADMIN-01 — Administration

Field	Type	Mandatory	Validation / source
Users & roles	CRUD	Yes	RBAC role assignment (NFR-SEC-01); from AD where available (INT-AD-01).
Stores & categories	CRUD	Yes	Master data for eligibility & reporting.
Reason codes	CRUD	Yes	Used by return intake (FS-RET-01).
Rule thresholds (500 / 5,000 THB, fraud N)	Edit	Yes	Drives approval routing (BRL-002/003) and fraud flagging (BRL-007).
Change audit (before/after)	System	—	Every change logged (NFR-AUDIT-01).

Pre-conditions: The user is authenticated with an administrator role. **Post-conditions:** Master data/thresholds persisted, RBAC applied to users, and every change written to the audit log. **Actors:** Administrator. **Main flow:** select entity → create/edit → validate → save → audit row written; new thresholds apply immediately. **Alt/exception:** non-admin denied (TR-ADMIN-02); invalid threshold rejected with message. **Uses status model:** configures the thresholds that drive approval routing and FRAUD_HOLD transitions in the shared model (§9.0). **BRL:** 002,003,007. **TR:** ADMIN-01..03. **US:** US-015.

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10 User Stories & Acceptance Criteria

Format: As a <role>, I want <action>, so that <value>. INVEST-aligned, grouped under Epics. Size: S/M/L (relative). Every **MUST** story has ≥1 **negative** Gherkin scenario (§10.1–10.2).

US	Epic	Story (As a… I want… so that…)	Pri	Size	TR links
US-001	EP-01 Customer Intake	As a **customer**, I want to submit a return online (web/LINE) selecting my order & items, so that I don't have to email or call.	MUST	M	RET-01,02,03
US-002	EP-01 Customer Intake	As a **customer**, I want a QR code to drop off at a store, so that staff find my return instantly.	COULD	S	RET-05
US-003	EP-01 Customer Intake	As a **customer**, I want to track my return status step-by-step, so that I know where my refund is.	MUST	S	NOTIF-01
US-004	EP-02 In-Store	As a **store staff**, I want to scan a receipt barcode to pull the order, so that I don't re-type anything.	MUST	M	STORE-01,02
US-005	EP-02 In-Store	As a **store staff**, I want to record item condition and photos, so that disposition and fraud checks are reliable.	MUST	S	STORE-03
US-006	EP-03 Approval	As **CS staff**, I want a prioritized approval queue, so that I can decide returns quickly.	MUST	M	APPR-01,03
US-007	EP-03 Approval	As a **branch manager**, I want to approve refunds > 5,000 THB, so that large refunds are controlled.	MUST	S	APPR-02
US-008	EP-03 Approval	As **Finance**, I want visibility/approval of every refund, so that audit and control are satisfied.	MUST	M	APPR-04
US-009	EP-04 Refund	As **Finance**, I want approved refunds posted to SAP automatically, so that I stop manual Friday re-keying.	MUST	L	REF-01,03
US-010	EP-04 Refund	As **Finance**, I want a daily refund report in Excel, so that I can reconcile.	SHOULD	S	DASH-04
US-011	EP-05 Disposition	As a **store staff**, I want to decide restock / return-to-supplier / write-off per item, so that goods don't pile up.	MUST	M	INV-01,03
US-012	EP-06 Notification	As a **customer**, I want SMS/LINE/email updates at each step, so that it feels like courier tracking.	MUST	M	NOTIF-01,02
US-013	EP-07 Dashboard	As an **executive**, I want a returns dashboard (volume, reason, store, lead time, fraud), so that I can manage performance.	MUST	M	DASH-01,02
US-014	EP-07 Dashboard	As a **CS fraud analyst**, I want serial returners flagged, so that I can investigate.	SHOULD	M	FRAUD-01
US-015	EP-08 Admin	As an **administrator**, I want to manage users, roles and thresholds, so that the system adapts without code changes.	MUST	M	ADMIN-01,02,03

Epic summary: EP-01 Customer Intake · EP-02 In-Store Processing · EP-03 Approval · EP-04 Refund · EP-05 Disposition · EP-06 Notification · EP-07 Dashboard & Fraud · EP-08 Administration — **15 stories across 8 epics.**

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10.1 Acceptance criteria (Gherkin) — EP-01 to EP-03 (incl. negative scenarios)

Story	Scenario (Given / When / Then)	Type
US-001	**Happy:** Given a customer with an order inside the return window, When they select an eligible item, add a reason and a photo and submit, Then a Return ID is issued and status = SUBMITTED and they receive a confirmation.	Positive
US-001	**Negative:** Given an item outside its category return window (BRL-001), When the customer tries to select it, Then the item is blocked with the reason "outside return window" and cannot be submitted.	Negative
US-001	**Negative:** Given a return without any photo, When the customer submits, Then submission is rejected with "photo required".	Negative
US-003	**Happy:** Given an existing return, When its status changes at each step, Then the customer sees a courier-style timeline.	Positive
US-003	**Negative:** Given a user with no matching return, When they open tracking, Then they see "no return found" and no other customer's data.	Negative
US-004	**Happy:** Given a valid receipt, When staff scan it on a tablet, Then the matching order and lines are retrieved in ≤ 2s.	Positive
US-004	**Negative:** Given an unreadable/unknown receipt, When staff scan, Then the system reports "order not found" and offers manual search (no crash).	Negative
US-005	**Happy:** Given an item being returned, When staff set a condition grade and add ≥1 photo, Then both are stored with the case.	Positive
US-005	**Negative:** Given no photo and no grade, When staff proceed, Then the item cannot advance.	Negative
US-006	**Happy:** Given cases awaiting review, When CS opens the queue, Then cases are ordered by SLA/age with timers.	Positive
US-006	**Negative:** Given a refund > 5,000 THB, When CS tries to approve without a manager, Then approval is blocked (BRL-003).	Negative
US-007	**Happy:** Given a >5,000 THB refund, When the manager approves, Then the decision, identity and timestamp are recorded and the refund is queued.	Positive
US-007	**Negative:** Given a refund of exactly 5,000 THB or less, When a non-manager attempts to use the manager-approval action, Then the action is rejected and the case stays in the CS/Finance queue.	Negative
US-007	**Negative:** Given a case already approved, When a manager re-submits an approval, Then the system blocks the duplicate approval (idempotent, no second refund).	Negative
US-008	**Happy:** Given any approved refund, When Finance opens the ledger, Then every refund is visible with full audit detail (BRL-004).	Positive
US-008	**Negative:** Given the <500 THB auto-approval branch, While OI-01 is unresolved, Then the branch is disabled and such refunds route to CS/Finance per current default.	Negative

> **OI-01 dependency:** US-006/US-008 scenarios that touch the <500 THB auto-approval branch cannot be finalised until the CS-vs-Finance conflict (OI-01) is decided. The acceptance above reflects the safe default (route to Finance) until then.

10.2 Acceptance criteria (Gherkin) — EP-04 to EP-08

Story	Scenario (Given / When / Then)	Type
US–009	**Happy:** Given an approved refund, When posted, Then a SAP document is created via TK-Connect, channel/ref stored, status = REFUNDED.	Positive
US–009	**Negative:** Given a refund already posted, When a duplicate post is attempted, Then it is rejected idempotently (BRL-014) — no double refund.	Negative
US–009	**Negative:** Given SAP is unavailable, When posting fails, Then the refund is marked POST_FAILED, retried, and raised in the daily reconciliation report.	Negative
US–010	**Happy:** Given refunds posted today, When Finance runs the daily export, Then an Excel file of todays refunds downloads and matches the dashboard.	Positive
US–011	**Happy:** Given a received item, When staff choose a disposition, Then the item is routed and status updated (BRL-009).	Positive
US–011	**Negative:** Given no disposition set, When staff try to close the return, Then closing is blocked until every item has a disposition.	Negative
US–012	**Happy:** Given a status change, When it occurs, Then a notification is sent on the customer's chosen channel in their language.	Positive
US–012	**Negative:** Given a channel fails, When sending fails, Then the system retries and, on final failure, falls back to email and logs it.	Negative
US–013	**Happy:** Given daily data, When an executive opens the dashboard, Then all KPIs render with date/store/category/channel filters.	Positive
US–013	**Negative:** Given a user without dashboard permission, When they access it, Then access is denied (RBAC).	Negative
US–014	**Happy:** Given a customer exceeding the return threshold, When evaluated, Then they are flagged and appear on the CS fraud list (BRL-007).	Positive
US–015	**Happy:** Given an admin changes a rule threshold, When saved, Then the change takes effect and is audit-logged.	Positive
US–015	**Negative:** Given a non-admin, When they attempt to change a threshold, Then the action is denied.	Negative

INVEST note: all stories are independent, negotiable, valuable, estimable, small enough to fit a sprint, and testable via the Gherkin above. SHOULD stories (US-010, US-014) may slip to a later sprint without breaking Must outcomes.

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11 Traceability Matrix — BR ↔ TR/NFR/INT ↔ FS ↔ US ↔ AC

Objective	Business rule(s)	TR / NFR / INT	FS	US	Acceptance
BR-01	BRL-011, BRL-014	TR-REF-01..04 · NFR-PERF-01 · INT-SAP-01 · INT-PAY-01	FS-REF-01	US-009	US-009 AC (post + idempotent + failover)
BR-02	BRL-011	TR-NOTIF-01,02 · TR-DASH-01 · INT-SMS/LINE/EMAIL	FS-NOTIF-01, FS-DASH-01	US-003, US-012	US-003/US-012 AC
BR-03	BRL-001,005,006,008,013	TR-RET-01..04 · TR-STORE-01..03 · TR-APPR-01..04 · TR-REF-01 · INT-POS-01	FS-RET-01, FS-STORE-01, FS-APPR-01	US-001, US-004, US-005, US-006, US-007, US-008	US-001/004/005/006/007/008 AC
BR-04	BRL-009	TR-INV-01..03	FS-INV-01	US-011	US-011 AC
BR-05	BRL-007	TR-FRAUD-01,02	FS-FRAUD-01	US-014	US-014 AC
BR-06	BRL-012	NFR-SCAL-01 · TR-STORE-01 (lookup perf)	FS-RET-01, FS-STORE-01	US-001, US-004	Load test at 4,200/day
BR-07	—	TR-DASH-01..04	FS-DASH-01	US-010, US-013	US-013/US-010 AC
BR-08	BRL-002, BRL-003, BRL-007	TR-ADMIN-01..03 · NFR-SEC-01 · NFR-AUDIT-01	FS-ADMIN-01	US-015	US-015 AC (RBAC + audit-logged config)
– (PDPA)	—	NFR-PRIV-01 · NFR-AUDIT-01 · NFR-SEC-01 · INT-AD-01	FS-ADMIN-01	US-008, US-015	US-008/US-015 AC
– (Usability)	—	NFR-USA-01 · TR-STORE-04	FS-STORE-01	US-004, US-005	≤5 taps usability test
– (Localisation)	—	NFR-LOC-01 · TR-RET-06	FS-NOTIF-01, FS-DASH-01	US-001, US-012	TH/EN toggle

Coverage check: every **MUST** TR/NFR traces to ≥1 BR and ≥1 US with acceptance criteria. **BR-08** was added so the ADMIN module (TR-ADMIN-01..03) traces to an objective; **TR-STORE-03** and **TR-APPR-02..04** now map to **BR-03** with US-005/US-006/US-007/US-008. Open dependencies flagged: BRL-002/004 → OI-01; BRL-001 → OI-02; INT-POS-01 → OI-08; INT-AD-01 → OI-07; BRL-006 → OI-10; BRL-010 → OI-05; NFR-DATA-01 → OI-06.

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12 Assumptions, Constraints & Dependencies

12.1 Assumptions (AS) — unverified facts with an owner

ID	Assumption	Impact if wrong	Owner
AS-01	Return volume sized at the **higher figure 25,000/month** (K.Somchai) vs 18,000 (K.Nok) (S-35).	Capacity/sizing/cost under-estimated if higher.	K.Arthit (confirm via POS/SAP)
AS-02	Historical data migration limited to **last 1 year** of Excel records (S-36).	More years = more effort/cost.	K.Arthit / K.Pim
AS-03	Photo/record retention = **7 years** pending audit confirmation (S-23).	Longer/shorter retention changes storage & compliance.	K.Pim (audit)
AS-04	Store-credit **+5% bonus** is a marketing idea, **not confirmed** — excluded until approved (S-13).	If included, changes refund calc & accounting.	Marketing / K.Somchai
AS-05	No native app; customer channel = web + **LINE OA** (S-09).	App demand = extra Phase-2 scope.	K.Somchai
AS-06	Refund target **≤3 business days (p95)** = competitor benchmark (S-05).	Stretch/relaxation changes SLA & design.	K.Somchai
AS-07	Complaint baseline **~2,300/quarter** is indicative, needs confirm (S-06).	BR-02 target baseline off.	K.Somchai
AS-08	Phase-1 budget **≤ 8M THB** (K.Somchai; Finance not fully agreed) (S-26).	Budget gap risk (see §14, OI-11).	K.Somchai / K.Pim
AS-09	Go-live **before 11 Nov 2027** (S-27).	Fixed date may be at risk (§14).	Steering group
AS-10	First release is a **pilot of 10 Bangkok branches** (S-28).	Rollout sequencing & effort.	K.Nok
AS-11	SAP ECC 6.0 exposes RFC usable **via TK-Connect** (no direct access) (S-14).	If no usable RFC, integration redesign.	K.Arthit
AS-12	POS (vendor TBD) can expose order lookup interface (S-15 / OI-08).	No API = slower/alternate integration.	K.Arthit
AS-13	SiamMart has AD available for SSO (S-37 / OI-07).	No AD = local accounts + weaker SSO.	K.Arthit
AS-14	Tablets for store processing are customer-provided / already available (S-10).	Hardware cost added to budget.	K.Arthit
AS-15	SMS gateway + LINE OA channels are procurable by TK (S-18).	Channel cost/fees (§14).	TK
AS-16	The payment gateway/bank for card & PromptPay refunds exposes an API usable via TK-Connect (exact gateway/vendor to be confirmed — INT-PAY-01) (S-13).	Gateway change/absence alters the refund integration & fees.	K.Pim / K.Arthit
AS-17	BR-02 target of **–50% refund complaints within 2 quarters** is a proposed customer target, **not sourced** — to be confirmed (S-06).	Target may need restating; does not affect build scope.	K.Somchai

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12.2 Constraints

Type	Constraint	Source
Budget	Phase-1 budget ≤ 8M THB (Finance not fully agreeing) — see AS-08/OI-11.	S-26
Deadline	Go-live before 11 Nov 2027.	S-27
Technology	SAP ECC 6.0 with RFC; must not be touched directly — integrate only via TK-Connect.	S-14
Operating model	TK to host & support; SiamMart IT has only 4 staff.	S-29
Licensing	Minimise expensive licences.	S-30
Regulation	PDPA compliance for all customer data and photos.	S-24
Usability	Must be very easy for older staff; TH/EN.	S-21,S-25
Scale	Handle 4–5× return peak after 11.11/12.12.	S-22

12.3 Dependencies

ID	Dependency	Needed from	By when / link
DEP-01	Return policy document (windows by category)	K.Nok	Before design — OI-02
DEP-02	SAP RFC access & sandbox via TK-Connect	K.Arthit	Before build — INT-SAP-01
DEP-03	POS vendor identification & API spec	K.Arthit	Before build — OI-08
DEP-04	AD/LDAP details for SSO	K.Arthit	Before build — OI-07
DEP-05	Payment gateway/bank API for card/PromptPay refunds	K.Pim / K.Arthit	Before refund build — INT-PAY-01
DEP-06	LINE OA channel & SMS account	K.Somchai / TK	Before notification build
DEP-07	Confirm which process owner (CS vs Store Ops) signs UAT	K.Somchai / K.Nok	Before UAT — OI-03
DEP-08	Store walk-through at Bangna branch (date TBC)	K.Nok	Requirements validation (S-41)

> The 8M budget and 11 Nov 2027 date are recorded as constraints but are **at risk** against Must-scope — see the Commercial Notes (§14) and OI-11.

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13 Open Issues & Conflicts

No stakeholder conflict is resolved by assumption. Each issue lists Position A, Position B, decision owner and due date. ⚑ **Blocking** = must be decided before design freeze.

ID	Title & both positions	Owner	Due
OI-01 ⚑	**<500 THB auto-approval vs all refunds via Finance.** **Position A (K.Somchai, Head of CS):** refunds under 500 THB should auto-approve for speed. **Position B (K.Pim, Finance):** every refund must pass through Finance, per audit — no auto-approval.	K.Somchai + K.Pim (escalate to COO / Head of Finance)	15 Nov 2026
OI-02 ⚑	**Return policy window by category.** 7 days? 14 days? electronics 7 / others 30? Policy doc not provided. **Position A:** single 7–14 day window. **Position B:** category-specific (electronics 7, other 30).	K.Nok (Store Ops) — provide policy doc	7 Nov 2026
OI-03	**Process owner: CS vs Store Ops.** Who owns and signs-off the end-to-end return process? **Position A:** Customer Service owns it. **Position B:** Store Operations owns it.	K.Somchai / K.Nok	7 Nov 2026
OI-04	**Return volume 18k vs 25k / month.** K.Nok ~18k; K.Somchai ~25k. Sized at 25k (AS-01) — needs data confirmation. **Position A:** ~18k/mo. **Position B:** ~25k/mo.	K.Arthit (pull POS/SAP data)	10 Nov 2026
OI-05	**Exchange (change-of-goods) in scope?** **Position A:** include exchange in Phase 1. **Position B:** refunds only; exchange later.	K.Somchai	15 Nov 2026
OI-06	**Photo/record retention period.** Audit "7 years"? not sure. **Position A:** 7 years. **Position B:** shorter operational retention.	K.Pim / K.Arthit	10 Nov 2026

Continued on next page →

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13.1 Open Issues (continued)

ID	Title & both positions	Owner	Due
OI-07	**SSO with SiamMart AD?** AD availability unconfirmed. **Position A:** integrate with AD. **Position B:** local accounts for Phase 1.	K.Arthit (IT)	10 Nov 2026
OI-08	**POS vendor:** "RetailPro" or in-house? K.Arthit not sure. **Position A:** RetailPro with API. **Position B:** in-house custom POS.	K.Arthit (IT)	10 Nov 2026
OI-09	**Marketplace (Shopee/Lazada) returns — Phase 2 confirm.** **Position A:** include in Phase 2. **Position B:** defer indefinitely / handle outside system.	K.Somchai	15 Nov 2026
OI-10	**Partial return allowed? (probably yes).** **Position A:** allow partial per line item. **Position B:** full-order returns only.	K.Nok / K.Somchai	7 Nov 2026
OI-11 ⚑	**Budget 8M THB vs Finance concerns.** K.Somchai says ≤8M; Finance "doesn't quite agree". **Position A:** hold to ≤8M by phasing. **Position B:** larger budget to cover Must scope.	K.Somchai / K.Pim	20 Nov 2026
OI-12	**Data migration scope: "1 year back enough".** Confirm exact cut-off & fields. **Position A:** 1 year only. **Position B:** more history.	K.Arthit / K.Pim	10 Nov 2026
OI-13	**Complaint baseline ~2,300/quarter — confirm number.** **Position A:** ~2,300. **Position B:** different figure.	K.Somchai	10 Nov 2026
OI-14	**Store-credit +5% bonus — confirm or drop.** **Position A:** offer 5% bonus. **Position B:** no bonus (marketing idea only).	Marketing / K.Somchai	15 Nov 2026

> **Blocking issues (design cannot freeze until decided):** OI-01 (approval model), OI-02 (eligibility window), OI-11 (budget). The rest are needed before build/UAT but do not block high-level design.

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14 Commercial Notes (INTERNAL — TK ONLY)

⚠ **This section is internal to TK. Remove it entirely before issuing any customer-facing copy of this document.**

14.1 Value sizing (why SiamMart should buy)

- **Refund lead time ~14 → ≤3 days:** tackles the #1 social complaint and a direct competitive gap (competitor 3 days).
- **~2,300 refund complaints/quarter** baseline — halving them protects brand and reduces CS rework.
- **25k returns/month** (sized high) — removing manual paper/Excel/LINE handling and the weekly SAP batch produces hard labour savings and auditability.
- **Returned-goods disposition** unlocks recovered inventory value (restock / supplier credit vs write-off).

14.2 Budget fit — Must scope likely exceeds the ≤ 8M THB phase-1 budget

Phase-1 **Must** scope is broad for 8M THB: 9 modules, two hard ERP/POS integrations (SAP ECC RFC + POS), PDPA-grade security, notifications across three channels, dashboards, plus MA hosting — for ~120 branches and 25k returns/month. TK assessment: **delivering every Must item with full integration + MA hosting inside 8M is high-risk.** Recommend one of:

- **Option A (recommended):** Phase 1 fixed at ~8M covering pilot 10 Bangkok stores + core Must (RET/STORE/APPR/REF/INV/NOTIF + basic DASH); defer FRAUD-advanced, QR (COULD), marketplace, exchange to Phase 2. Protects budget and the Nov 2027 date.
- **Option B:** retain full Must incl. fraud + advanced dashboard; requires budget uplift (see OI-11) — present trade-off to Finance.

14.3 Hidden / recurring costs to budget explicitly

Cost item	Note
SMS + LINE OA messaging fees	Per-message cost at 25k returns × multiple steps × channels — recurring, scales with volume/peak.
TK-Connect middleware (SAP/POS)	Integration layer to keep SAP untouched; build + run cost.
Data migration (1 year Excel)	Cleansing the "RETURN_MASTER_v7" data is effort-heavy; scope creep if more years (OI-12).
Hosting + MA support	Customer wants TK to host & support (only 4 IT staff) — recurring annual fee; avoid expensive licences per S-30.
Payment gateway/bank fees	Per-refund processing fees on card/PromptPay.
Store tablets (~120 branches, if not owned)	Hardware capex if AS-14 is wrong; pilot 10 first.
Hypercare post go-live	Especially around 11.11/12.12 peak.

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14.4 Deadline risk (go-live before 11 Nov 2027)

The fixed date is aggressive because it coincides with the 11.11 peak and sits behind **unresolved integrations** (POS vendor OI-08, AD OI-07) and **unresolved rules** (approval OI-01, policy OI-02). If OI-01/OI-02/OI-08 are not closed by early Nov 2026, the date is at risk. Recommend a hard decision gate and a pilot-first (10 stores) de-risking plan.

14.5 Scope-creep guard

Everything the customer said "maybe / phase 2 / nice to have" is already **Out of Scope** with routing (§3.2): marketplace returns, native app, exchange, AI photo check, store-credit bonus. Any request to pull one of these into Phase 1 must go through a formal **Change Request** with budget/date impact — no informal absorption.

14.6 Upsell / future opportunities

- **Phase 2:** Shopee/Lazada marketplace returns, exchange workflow, mobile app, store-credit loyalty programme (with bonus).
- **AI item-condition photo check** — noted as "nice to have"; strong Phase-2 upsell.
- **Advanced fraud analytics** (ML scoring beyond the configurable threshold).
- **MA / managed-service** multi-year contract with TK hosting — recurring revenue.
- **BI/analytics** expansion onto the returns data mart.

14.7 Commercial risks register

Risk	Impact	Mitigation
Budget not agreed by Finance (OI-11)	Deal stall / descope	Present Option A phased plan
Unknown POS vendor (OI-08)	Integration effort/cost blowout	Discovery before fixed price
Approval-model conflict (OI-01)	Rework of core workflow	Gate design on decision
Fixed go-live vs 11.11 peak	Quality/date trade-off	Pilot-first, phased rollout
Data migration underestimated (OI-12)	Effort overrun	Cap at 1 year, fixed field set

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15 Sign-off

By signing below, the parties confirm that the requirements in this package (sections 1–13) accurately capture the agreed Phase-1 scope for the SiamMart Returns project, subject to the open issues in section 13 being resolved. Section 14 (Commercial Notes) is internal to TK and is not part of the customer sign-off.

Name	Role	Signature	Date
K.Somchai	Head of Customer Service (SiamMart)		
K.Pim	Finance (SiamMart)		
K.Arthit	IT (SiamMart)		
K.Nok	Store Operations (SiamMart)		
	TK Company — Delivery Lead		

> **Sign-off is conditional** on decisions for the blocking OIs: `OI-01` (approval model), `OI-02` (return window), `OI-11` (budget). The remaining OIs must be closed before build/UAT.

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Appendix A Glossary

Term	Definition
AS / OI / BRL / TR / NFR / INT / FS / US / EP	Assumption · Open Issue · Business Rule · Technical Requirement · Non-Functional Requirement · Integration Requirement · Functional Spec · User Story · Epic.
MoSCoW	Prioritisation: Must / Should / Could / Won't (this phase).
p95	95th percentile — 95% of cases are at or below this value.
RTO / RPO	Recovery Time Objective / Recovery Point Objective.
Idempotent	An operation that produces the same result even if repeated — prevents double refunds.
TK-Connect	TK integration layer through which SAP/POS/marketplace traffic flows (no direct DB access).
Partial return	Returning only some line items of an order.
Disposition	Final routing of a returned item: restock / return-to-supplier / write-off.
Store credit	Refund issued as store value instead of cash to the original payment method.
LINE OA	LINE Official Account — SiamMart's customer messaging channel and a return intake channel.
PDPA	Thailand Personal Data Protection Act — governs consent, retention, masking and erasure of personal data.
PromptPay	Thailand national interbank payment rail used for cash-equivalent refunds.
MA	Maintenance Agreement — ongoing support contract.
DC	Distribution Centre.
SAP ECC 6.0	SiamMart's ERP/finance system; refunds are posted to it via RFC through TK-Connect.
UAT	User Acceptance Testing.
สาขา (sākhā)	Branch / store.
คืนของ (khuen khong)	Return goods / product return.
การคืนเงิน (kān khuen ngern)	Refund.
ใบเสร็จ (baiset)	Receipt.
บางนา (Bāngnā)	Bangna — a Bangkok district; site of the planned process walk-through branch.
พ.ศ. (Phō.Sō.)	Buddhist Era year (Thai calendar); 2027 CE ≈ 2570 พ.ศ.
RETURN_MASTER_v7_final_FINAL.xlsx	The ad-hoc central Excel file currently used by CS to track online returns (to be replaced).

— End of document — **TK-BA-SMR-REQ-01 · v1.0 · TK Company – Confidential**

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